CS4928 Theory and Practice of Software Design

2026/27 Group Project Brief and Marking Rubric

This project is designed for contemporary software development where generative AI tools are normal development tools. You may use ChatGPT or another GenAI tool throughout the project unless a specific in-person activity says otherwise. The assessment is not based on proving who typed individual lines of code. It is based on the quality, traceability and verification of the team's engineering and empirical work, together with the separate individual in-person examination.

1. Assessment

Component	Weight	How it is assessed
Individual in-person MCQ examination	40%	Scenario/code/UML/design-decision questions under controlled conditions.
Group empirical software-design project	60%	Common published rubric applied to the final evidence trail, repository/workbook, event responses and concise final synthesis.

There is no project viva and no peer-mark allocation. The project is marked against the published criteria; where groups will be assessed progressively during lab sessions; groups are not ranked against one another. Note: you are allowed to bring your own laptops to labs if you are not interested in using the provided lab's systems.

2. What the project is

Each group will work on one recent empirical software-engineering study and its public research artefact/repository from the released project-pack list. Your job is not to reproduce the whole paper. "You will reproduce one bounded part of the study, audit the evidence, extend or transfer the method to new material, make a justifBrief andied software-design/refactoring/modernisation decision, and respond to two later Change Events".

1. Groups: normally 4 students; maximum 5.
2. Each project pack identifies the language/technology of its repository or artefact. Choose a pack whose existing stack you are prepared to work with and normally use that stack rather than rewriting the project in another language. Most primary packs use Java or a Java-selected subset, while a small number naturally use another route.
3. Normally use two research questions (RQs).
   RQ1 should address the bounded reproduction.
   RQ2 should address transfer/extension and the software-design/modernisation question.
   A third RQ is optional only when it clearly adds value.
4. A failed reproduction, negative result or decision not to introduce a pattern can earn a high mark when you can demonstrate that the employed method, evidence and interpretation are strong.
5. Do not manufacture a result to match the paper (this is important as in RQ1, you are supposed to replicate the (part of) paper as is irrespective of what results you achieve). Record what the evidence shows.

3. Project stages and required freezes

When	Stage	Minimum record
Week 5 – M1	Study + RQs + preregistration	Record/freeze RQs, bounded target, sample IDs, method, metric/decision rule, expected result and what would count as non-reproduction. Tag: M1_PLAN.
Week 7 – M2	Bounded reproduction + audit	Complete the bounded reproduction or document a defensible infrastructure limitation using the contingency route. Record raw/result evidence and key deviation/threat. Tag: M2_REPRO and PRE_EVENT_A.
Week 8	Change Event A	Read only after PRE_EVENT_A is frozen. Record your prediction before executing the event analysis.

When	Stage	Minimum record
Week 9 – M4	Design / refactoring / modernisation	Identify the concrete design pressure, compare at least two realistic responses including no structural change where appropriate, implement/analyse the smallest justified intervention, and verify it. Tag: PRE_EVENT_B.
Week 10	Change Event B	Respond to the new evolution/architecture/trade-off pressure from the frozen M4 state.
Week 11	Final	Complete final evidence, reproducibility notes and concise synthesis. Tag: FINAL. All assessed project work finishes by the end of Week 11.

4. The shared group workbook

Your group will use one shared class workbook, **Student Tracker.xlsx**, throughout the project. You only update the cells belonging to your own group. This workbook is part of the assessed project evidence and helps you prepare for the project checks carried out during the labs.

The workbook contains four sheets:

- **Group Allocation:** enter your group members' names only by Week 3. Project, paper and repository information is already provided.
- **RQs and Milestones:** maintain your group's brief project record at the required stages:
  - **Week 5:** RQ1 and RQ2, bounded sample/cases, method/decision rule and expected result before execution.
  - **Week 7:** bounded reproduction result and the evidence/output that supports it.
  - **Week 8:** Event A prediction before carrying out the event, followed by the result/decision.
  - **Week 10:** concrete design/modernisation problem, alternatives considered, chosen intervention or no-change decision, and verification evidence.
  - **Week 11:** Event B prediction and response, final repository/tag, concise final conclusion and main limitation/threat.
- **Week 9 Exercise – Cross-Team Review:** complete the review exercise for one other group present in your scheduled lab and record your response to the review received by your group. You must need to select (and insert the information of) the group you will be reviewing in Week 8. Always choose a group attending the same lab as you.
- **Rubric:** provided for reference only. **Students do not enter marks, evidence levels or coordinator information on this sheet.**

Keep entries short and evidence-linked. Where possible, point to a repository commit/tag, script, test, raw output, table/figure or other locatable evidence rather than copying large amounts of material into the workbook.

Do not rewrite an earlier prediction or RQ after seeing the result. If it genuinely changes, retain the earlier record and explain the change briefly.

The shared workbook will be **frozen by staff at the end of Week 11 and used as part of the project evidence during marking**. You do not need to submit a separate copy of this workbook unless specifically instructed.

5. AI use

AI use is permitted (and sometimes expected in the project). You remain responsible for everything accepted into the project. "*Treat AI as a junior engineer/research assistant whose output must be checked against the actual paper, repository, code, tests and data*".

- AI may help read code, propose patterns/refactorings, write scripts/tests, explain results and review a diff.
- AI output is not evidence by itself. Evidence is: **the source code/data, a reproducible computation, a test/verification result, or a traceable design decision.**
- My suggestion: *Do not claim a pattern, smell, causal explanation, metric improvement or successful refactoring because an AI tool said so. Verify it.*

- You may accept, modify or reject AI advice. Good rejection based on evidence is a valid engineering outcome.

6. Final submission through Brightspace Links (As it is a group Project, No Exception/Late-Submission Allowed)

- Final repository/workspace and FINAL tag/commit. Must be submitted in Week 11.
- Completed group record from the shared student workbook. Must be frozen in Week 11; no need to submit.
- Runnable/traceable scripts, commands, raw/intermediate results and tests needed to support the claimed results. Must be submitted in Week 12.
- A ***concise*** final synthesis PDF, IEEE conference format, maximum 6 pages excluding references, covering RQs, method, results, design decision, both Change Events, limitations and conclusion. Point to evidence rather than duplicating large outputs. Must be submitted in Week 12.
- Only One Member of the group will do all these submissions. Where required, submissions must be named after group ID. Remember, any additional submissions will be disregarded and will not be assessed.

7. Universal project rubric – 60 marks

Criterion	Marks	What strong evidence looks like
RQ and empirical design	8	Clear bounded RQs; correct unit of analysis; preregistered method/decision rule; appropriate scope; changes to RQs are traceable rather than rewritten after results.
Bounded reproduction and methodological accuracy	10	Method follows the relevant paper component closely enough to answer RQ1; deviations are explicit and justified; negative/non-reproduction is analysed rather than hidden.
Evidence quality and reproducibility	10	Claims point to locatable raw/intermediate evidence, code, tags, scripts and outputs; another technically competent person could audit/re-run the bounded work.
Software design / refactoring / modernisation decision	12	Concrete design pressure identified; alternatives compared; decision uses module concepts appropriately; intervention/no-change is proportionate; behaviour/effects are verified.
Response to Change Events A and B	10	Pre-event state and prediction are traceable; team responds to new evidence/constraints rather than defending the old decision; evolution impact and trade-offs are evidenced.
AI-assisted engineering judgement and verification	5	AI is used critically; important AI suggestions are checked against code/data/tests; incorrect or over-engineered output is identified; material AI influence is recorded concisely.
Final synthesis, limitations and traceability	5	Concise coherent answer to the RQs; limitations/threats are credible; conclusions do not exceed evidence; final claims link back to the project record and artefacts.

Evidence levels used during marking: 5 = excellent/fully convincing; 4 = strong with minor gaps; 3 = competent/serviceable; 2 = partial/inconsistent; 1 = minimal; 0 = absent/not defensible. The criterion mark is the published weight multiplied by the evidence level / 5. The rubric is criterion-referenced; there is no curve.

8. Frequently asked questions (Please consult here before submitting your queries)

Is ChatGPT allowed? Yes. It is not a misconduct issue when used within this brief. You remain responsible for verification and accurate reporting.

Must our result match the paper? No. A failed(different results)/partial reproduction is a result if the method and evidence are sound.

Must we reproduce the complete paper? No. Each project pack defines a bounded target. Do not expand scope unnecessarily.

Can we change our RQs? Yes when evidence or a Change Event genuinely requires it. Keep the earlier wording visible and record why/when it changed.

Are all projects equally difficult? Different studies use different artefacts, but all are assessed using the same evidence standards and bounded scope. You are not compared with another group's scientific result.

What if a research artefact does not build? First document the exact failure and check the pack/Staff-approved contingency. You will not be penalised for a genuine external infrastructure failure, but you must still complete an auditable bounded analysis using the available data/output/fallback route.

What does the lab coordinator record? At scheduled checkpoints the coordinator records whether required evidence can be located and may ask one short question grounded in your own current project. These checks are evidence for the final rubric; they are not separate engagement or attendance marks.

Do we submit prompt logs? No. Record only brief material AI contributions when they affected an engineering/research decision.

Why do we freeze stages? To preserve what you believed and designed before later results/requirements were known, rather than reconstructing a perfect story afterwards.

If this document, in combination with the worksheets shared, does not answer your question, only then revert back with a question. Please ask questions to the right person: lecturer vs lab coordinator.